AI Agent Discovery

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Internal

SHOULD | SHOULD NOT

www.example.com is the label for a web site and is used to find it.

- DNS is not used to communicate with the web site, nor get the web site content.

index._agents.example.com could be the label to find an organization's AI agents (in this case, a registry).

- DNS is not used to communicate with the index of agents, nor get the details of all the agents.

DNS used as stable beachhead to find well-known agents, and well-known registries of agents.

Internal

SHOULD | SHOULD NOT

The DNS used to express full agent capabilities.

Enumerate the DNS to search for a best-suited agent for a task or all agents within a domain, OR all domains with agents supporting X capability.

Be used solely as a trust, verification, or identity-related mechanism.

Create a new RRType, or modify RDATA logic to send all records in a zone.

Internal

SCOPE

The client knows the agent it wants to interact with:

- The label (eg copilot)
- The domain / owner (eg microsoft.com)

The listed agent is assumed:

- Trusted and authenticated, client authorization validated etc (pre-discovery)
- Application protocols exist for all data that isn't DNS-expressable (post-discovery)

Internal

DRAFT

Discovery status:

- Known service & domain: DNS
- Known service | domain: index
- Known service NOR domain: registry / search-engine

Record status:

- _agents.example.com
- SVCB w/ params for agent/application-specific metadata & model card URI
- TXT fallback for DNS that don't support SVCB

Problem statement:

https://datatracker.ietf.org/doc/draft-mozley-aidiscovery/

https://github.com/danielkinguk/discovery/blob/main/README.md#problem-statement-and-work-scope

Internal

"SUPPORTERS"

Existing similar DNS-based AI discovery I-Ds (beyond ours):

https://datatracker.ietf.org/doc/draft-nemethi-aid-agent-identity-discovery/

https://datatracker.ietf.org/doc/draft-cui-dns-native-agent-naming-resolution/

https://datatracker.ietf.org/doc/draft-liu-cats-dns-service-discovery/

CN NIC side meeting presentation (waiting for link to slides / drafts)

Complementary drafts:

"Use TXT RR to list payment services" https://datatracker.ietf.org/doc/draft-jeftovic-x402-dns-discovery/

"The index to consolidate AI DNS records" https://ad.jaime.win/draft-jimenez-ad-agent-directory

"Combine index and DNS" https://www.ietf.org/archive/id/draft-narajala-ans-00.html

Internal

QUESTIONS

Internal

COMMON OBJECTIONS

Why not /well-known/?

- Out of band PKI to validate endpoint association
- Knowing HTTPS origin first + querying DNS to get there
- Caching semantics for CDN / intermediaries may fragment depending on metadata support... outages become single points of failure for entire discovery
- Fragment observability and policy controls between whichever options each HTTP origin implements, standardized view into what was discovered and by whom more difficult (eg we know chat._example.com from IP xxx.xxx.xxx.xxx is an AI workload)

… There will need to be registry / HTTP to solve for the use cases when you don't know the exact agent, our argument is most times enterprises through trust relationships will know the agent for the task

Internal